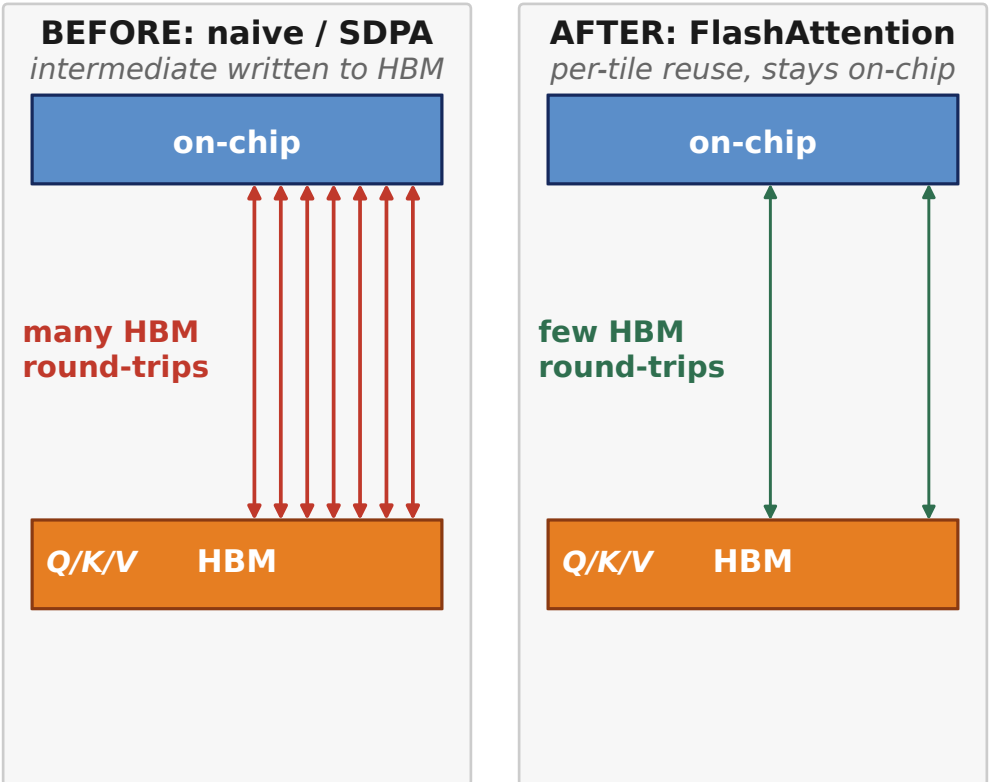
FlashAttention: the same math, far less HBM traffic



$O(L^2)$ HBM traffic:
every query re-reads
the K/V row-block

K/V reused per tile —
no single global K/V pass;
no L×L intermediate at HBM

same $Q \cdot K^T \cdot V$ math and same result —
the gain is fewer HBM round-trips, not cheaper attention

K/V on-chip reuse is per-tile (per Q-block), not a single global K/V pass.
I/O bound: FlashAttention — Tri Dao et al., arXiv:2205.14135.